The coordinate automorphism groups of the 120-coordinate graph and difference codes

Bounded proof supplement for the L050 reconstruction

The conclusions below are the automorphism assertions in Leemans–Rodrigues [1, Proposition 2 and its proof, printed p. 246]. We supply a complete bridge for the actual graph code and its row-difference subcode. The previously checked codes of dimensions 16 and 15 are different spaces; their minimum-support certificate is not used for either upper bound here. Automorphisms throughout are coordinate permutations, not arbitrary monomial transformations.

Fix the retained ordered Steiner system $\mathcal{W} = S(4, 7, 23)$. Its 120 heptads avoiding two fixed points are the coordinates. Let A be the binary adjacency matrix joining distinct heptads precisely when their intersection has size three. Thus the graph Γ has parameters $(120, 77, 52, 44)$; it is the degree-77 graph in [1], not its degree-42 complement. Work over $\mathbb{F}_3$ and set

$$C = \langle A_i : 1 \le i \le 120 \rangle, \qquad S = \langle A_i - A_1 : 2 \le i \le 120 \rangle.$$

The retained exact bases have dimensions 21 and 20. The completed enumeration [2], on these exact bases, gives minimum weight 30 and exactly 112 words of that weight for each code. The present replay binds those results and bases; it does not repeat their exhaustive enumeration.

Theorem 1. *For these actual coordinate spaces,*

$$\mathrm{Aut}(C) = \mathrm{Aut}(S) = \mathrm{Aut}(\Gamma) = G, \qquad |G| = 80640.$$

Here G is the group on the literal four coordinate generators retained in [3]. Its previously checked abstract identification is $L_3(4) : 2^2$. The equality and order in this theorem have the separate elementary certificate below; the abstract identification is retained from [3].

Lemma 2 (Recovering a graph from its code)**.** *For this matrix, $E = J - A$ is the orthogonal projector onto C. Consequently every coordinate permutation preserving C preserves A.*

Proof. The strongly regular parameters give $A^2 = 33I + 8A + 44J$, hence $A^2 = -A - J$ over $\mathbb{F}_3$. Also $AJ = JA = 2J$ and $J^2 = 0$. Therefore

$$E^2 = J^2 - JA - AJ + A^2 = J - A = E, \qquad EA = A.$$

Every row of J lies in C: the sum of the rows of A is $77\mathbf{1} = 2\mathbf{1}$. Thus $\operatorname{im} E \subseteq C$, and $EA = A$ gives the reverse inclusion (using symmetry to identify row and column spaces). Since E is symmetric and idempotent, its kernel is $C^\perp$ and its image is C, so $C \cap C^\perp = 0$. This determines the projector uniquely. A coordinate permutation preserving C also preserves the standard dot product and hence $C^\perp$; conjugation must fix E. It fixes J, so it fixes $A = J - E$. □

Lemma 3 (Recovering the graph from the difference code)**.** *The collection of supports of the minimum words of S determines A.*

Proof. The literal generators of G preserve A, hence preserve both C and S. Indeed, a permuted row difference is again a difference of two rows. The exact replay verifies a supplied word w lies in the actual S and has weight 30. Its orbit under these generators consists of 112 distinct words. The retained complete minimum-word count is 112, so this orbit is all the minimum words of S. It is closed under negation and has exactly 56 distinct supports. No orbit size is substituted for an unproved completeness assertion: that assertion uses the retained exhaustive count.

Let $\mathcal{B}$ be these 56 supports and put

$$t_{ij} = |\{B \in \mathcal{B} : i, j \in B\}|.$$

The replay constructs every orbit word from w, then computes every entry of this integer incidence Gram matrix. It gives $t_{ii} = 14$, and, on the 7140 unordered distinct pairs, the complete table is

A_{ij}	t_{ij}	Number of pairs	Interpretation
0	4	2520	nonedge
1	2	3360	edge
1	6	1260	edge

Thus, for $i \neq j$, $A_{ij} = 1$ if and only if $t_{ij} \neq 4$. Any coordinate permutation preserving S preserves Hamming weight and therefore permutes $\mathcal{B}$; it preserves every t_{ij} and hence A. □

Lemma 4 (An elementary full graph bound)**.** *The retained four permutations generate all of* $\mathrm{Aut}(\Gamma)$.

Proof. First the replay verifies that each is a permutation of all 120 coordinates preserving every adjacency entry. Closing the generated permutations under composition gives exactly 80640 distinct elements; only exact finite permutation operations are used.

For the reverse inequality, color refinement repeatedly replaces the color of a vertex by its old color together with the multiset of colors of its neighbors. Each refinement partition is preserved by every automorphism preserving the previous colors. Begin with one color. After refinement choose the first vertex in the smallest nonsingleton cell (lexicographic ties), give it a fresh color, and refine again. The following are the actual chosen vertices and sizes of their cells just before individualization; vertices in this table are one-based.

Step	Chosen vertex	Cell size	Previously fixed vertices
1	1	120	none
2	5	42	1
3	2	8	1,5
4	3	2	1,5,2

After the last refinement all 120 cells are singletons, so the pointwise stabilizer of these four vertices is trivial. At each step the orbit under the preceding pointwise stabilizer is contained in the displayed cell. The stabilizer-chain formula therefore gives $|\mathrm{Aut}(\Gamma)| \leq 120 \cdot 42 \cdot 8 \cdot 2 = 80640$. This matches the subgroup lower bound. □

Proof of Theorem 1. Every graph automorphism permutes rows of A and therefore preserves C and S. Lemmas 2 and 3 give the reverse inclusions. Lemma 4 identifies the common permutation group exactly. □

Exact certificate and retained computation. The standalone `check.py` imports only the unchanged full Steiner constructor from [3], besides the Python standard library. It constructs the 120-coordinate graph, checks its parameters, rederives both row spaces, verifies the projector, computes the 112-word orbit and all support incidences, performs graph refinement, and closes the explicit group. No GAP, graph-isomorphism package or numerical approximation is required by this new replay. Ten deliberate mutations check incomplete counts, a wrong seed, bad permutations, an insufficient subgroup, a wrong refinement bound, an incomplete orbit, the graph complement and confusion of the two code dimensions.

The seed was located by a bounded search through coefficient supports of size at most five in the retained 21-row basis (758451 messages). That search is discovery only. The certificate checks the supplied word directly in S and needs no sparse-search program. In contrast, the complete earlier word enumeration is a genuine retained premise for Lemma 3: all 3^{20} messages were accounted for, and the bound result gives exactly 112 words of weight 30. Both prior full enumeration vectors, their full bases, the whole original enumeration program, its frozen receipt and method-review manifest are included with exact hashes. The complete upstream computation remains a declared input; its runtime is not disguised as part of this small replay. The C projector argument does not need a minimum-word enumeration.

Scope and attribution. The source gives these group assertions; no new theorem priority or conjecture credit is claimed. This supplement establishes the two particular automorphism interfaces needed in the reconstruction of Proposition 2. It does not independently recover every L050 statement, establish the entire module lattice, claim a monomial automorphism group, add any Lean declaration, or infer a numerical compression factor. In particular it does not infer these assertions from the different N_{16} and K_{15} codes in Proposition 3. The group isomorphism notation is retained from the distinctly reviewed explicit group identification [3].

References

[1] D. Leemans and B. G. Rodrigues, *Linear codes with complementary duals from some strongly regular subgraphs of the McLaughlin graph*, exact L050 primary edition, Proposition 2 and proof, printed p. 246. The packet source map binds the complete eleven-page PDF and the graph-definition context on printed p. 243.

[2] Root-authored completed ternary enumeration, `completed_enumeration` (under the exact path recorded in the source map), exact 120 adjacency and difference bases/results. The method has a distinct bounded review in the bound enumeration method-review packet.

[3] Rewrite-termination, *Two ternary codes from flags and Fano subplanes*, `topic_LCD_120_modules`; distinct root review `topic_LCD_120_root_review`. Only the unchanged construction, literal group and its accepted identification are retained here, not the different code upper-bound argument.